

Dog Food

MATH 213

Group 1

Dog Food. An article in the February 20, 2009 issue of *Science* described a (lighthearted and informal) study in which 18 subjects (the author's New Year's Eve guests) were presented with six unlabeled blended meat products. One of these products was *Newman's Own* dog food, prepared with a food processor to have the texture and appearance of a liver mousse, and the other five were various gourmet mousse products. After tasting all six products, subjects were asked to identify which was their least favorite on the basis of taste. The research question is whether subjects are more likely to identify dog food as their least favorite than if they were making selections at random.

1. Who or what are the cases?

The 18 subjects (author's New Year's Eve guest)

2. What variable was recorded in this "study"? Is it categorical or numerical?

Least favorite product - categorical

3. Will your test involve a mean or a proportion? What notation will you use in the hypothesis test?

Proportion - p

It turns out that 13 out of 18 of the study subjects selected the dog food as their least favorite.

We are interested in testing the skeptic's theory that the subjects were in fact selecting their "least favorite" at random.

4. What proportion of the time would you expect the dog food to be selected if indeed the subjects were selecting at random? Calculate the proportion as a decimal.

$$\frac{1}{6} = 0.1667$$

5. Write out your null and alternate hypotheses, both in words and using correct notation.

H₀: the dog food did not affect whether it was a subjects least favorite or not. $P = 0.1667$

H_a: the dog food did have an effect if it was the subjects least favorite or not. $P > 0.1667$

6. Describe how you might use a spinner divided into 6 equal slices (see here) to simulate the null model where the subjects are selecting at random. Include enough detail so that someone could read your description and know all of the steps to carry out to make the analysis. (*Hint: What would the numbers on the spinner represent? What does each spin represent? How many spins? Etc.*) (I am asking you to explain this procedure to help you think through what is going on in the One-Proportion applet.)

The numbers on the spinner would represent each of the 6 products and each spin of the spinner would represent a subjects choice. There would be 18 spins because in the experiment there are 18 people.

7. We will use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the dog food study according to the **null model** proposed by the skeptic, where the subjects are selecting at random.

7a.What will you use for the probability of success in this situation?

% -> 0.1667

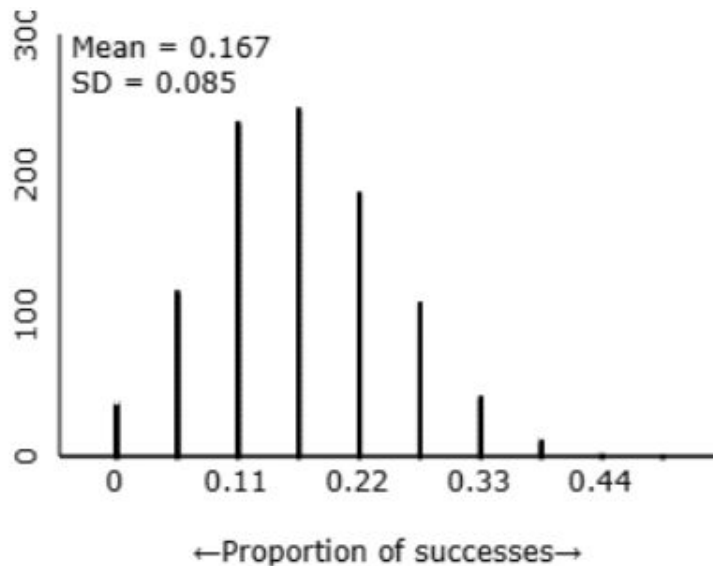
7b. What will you put in for the number of tosses?

18

7c. How many repetitions?

1000

8. Perform the simulation. Copy and paste the histogram you get here.



9. Describe what is going on in the graph in the usual way.

The histogram shows 1000 simulation of the dog food experiment, assuming that each participant choose their least favorite food at random.

The average proportion who chose dog food was .167% with a deviation to 0.085% choosing dog food. The graph is slightly skewed right.

10. What conclusions can you draw from your simulation? As part of your answer, make sure to calculate an appropriate p-value and use it to support your conclusions.

Since the experiment had a proportion of 13/18 or 0.722 of people who choose dog food as their least favorite, we would look at the number of simulations that are more extreme than 0.722.

When looking at the graph we can see that there are no simulations that have this proportion. Meaning that it is unlikely

Group 2

Dog Food. An article in the February 20, 2009 issue of *Science* described a (lighthearted and informal) study in which 18 subjects (the author's New Year's Eve guests) were presented with six unlabeled blended meat products. One of these products was *Newman's Own* dog food, prepared with a food processor to have the texture and appearance of a liver mousse, and the other five were various gourmet mousse products. After tasting all six products, subjects were asked to identify which was their least favorite on the basis of taste. The research question is whether subjects are more likely to identify dog food as their least favorite than if they were making selections at random.

1. Who or what are the cases?
 - The 18 people
2. What variable was recorded in this “study”? Is it categorical or numerical?
 - Least favorite product – categorical
3. Will your test involve a mean or a proportion? What notation will you use in the hypothesis test?
 - Proportion – p

It turns out that 13 out of 18 of the study subjects selected the dog food as their least favorite.

We are interested in testing the skeptic's theory that the subjects were in fact selecting their "least favorite" at random.

4. What proportion of the time would you expect the dog food to be selected if indeed the subjects were selecting at random? Calculate the proportion as a decimal.

$\frac{1}{6}$ choices = 0.1667

5. Write out your null and alternate hypotheses, both in words and using correct notation.

H_0 : The subjects choose their least favorite product at random. $p = 0.1667$

H_a : The subjects tend to choose dog food as their least favorite??? $p > 0.1667$

6. Describe how you might use a spinner divided into 6 equal slices (see here) to simulate the null model where the subjects are selecting at random. Include enough detail so that someone could read your description and know all of the steps to carry out to make the analysis. (*Hint: What would the numbers on the spinner represent? What does each spin represent? How many spins? Etc.*) (I am asking you to explain this procedure to help you think through what is going on in the One-Proportion applet.)

The numbers on the spinner would be each of the 6 products (including the dog food). Each spin represents a choice that a subject made. Because there were 18 subjects in the actual experiment, we would spin the spinner 18 times. This represents one simulation of the experiment under the assumption of the null hypothesis (that the subjects chose their least favorite product at random).

7. We will use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the dog food study according to the **null model** proposed by the skeptic, where the subjects are selecting at random.

7a. What will you use for the probability of success in this situation?

$\frac{1}{6}$ or 0.1667

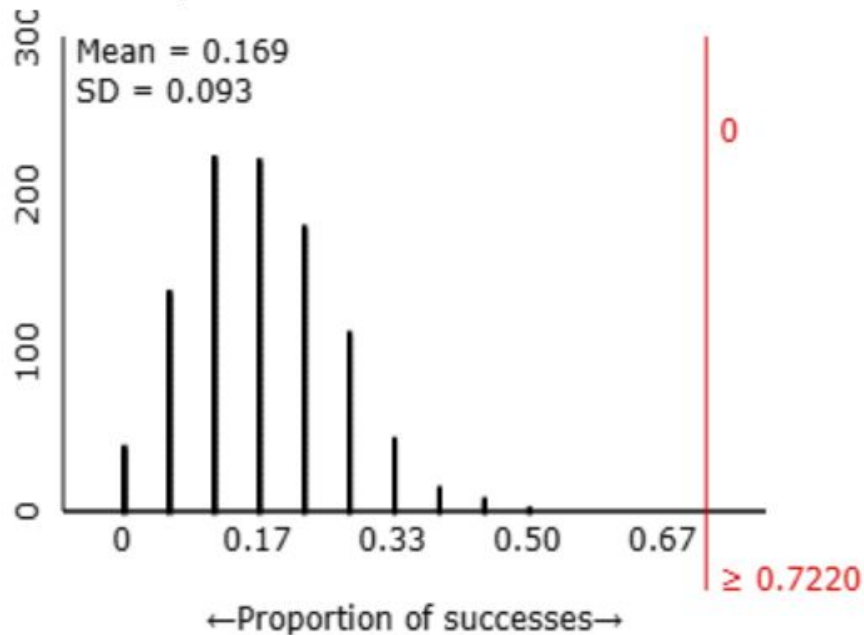
7b. What will you put in for the number of tosses?

18

7c. How many repetitions?

1000

8. Perform the simulation. Copy and paste the histogram you get here.



9. Describe what is going on in the graph in the usual way.

This histogram shows 1000 simulations of the experiment under the assumption that subjects choose their least favorite product at random. The average proportion who chooses dog food is 0.169, give or take 0.093, with some right skew.

10. What conclusions can you draw from your simulation? As part of your answer, make sure to calculate an appropriate p-value and use it to support your conclusions.

Because the actual experiment had a proportion of 13/18, or 0.722, who chose dog food as their least favorite, we look at the number of simulations (under the null hypothesis) that are as or more extreme than this value.

As it turns out, there were no (0) simulations who had this proportion – therefore, it is very unlikely that the results from the experiment were due to random chance/choice.

Group 3

Dog Food. An article in the February 20, 2009 issue of *Science* described a (lighthearted and informal) study in which 18 subjects (the author's New Year's Eve guests) were presented with six unlabeled blended meat products. One of these products was *Newman's Own* dog food, prepared with a food processor to have the texture and appearance of a liver mousse, and the other five were various gourmet mousse products. After tasting all six products, subjects were asked to identify which was their least favorite on the basis of taste. The research question is whether subjects are more likely to identify dog food as their least favorite than if they were making selections at random.

1. Who or what are the cases?

18 New years eve guests

2. What variable was recorded in this "study"? Is it categorical or numerical?

categorical

3. Will your test involve a mean or a proportion? What notation will you use in the hypothesis test?

Proportion, p

It turns out that 13 out of 18 of the study subjects selected the dog food as their least favorite.

We are interested in testing the skeptic's theory that the subjects were in fact selecting their "least favorite" at random.

4. What proportion of the time would you expect the dog food to be selected if indeed the subjects were selecting at random? Calculate the proportion as a decimal.

$$13/18 = .7222$$

5. Write out your null and alternate hypotheses, both in words and using correct notation.

H_0 : $P_{\text{dog food}} = 1/6$. The subjects randomly choose their least favorite product

H_a : $P_{\text{dog food}} > 1/6$. The subjects tend to choose dog food as their least favorite

Malec: be sure to write this with the notation too, which will force you to be more precise with your thinking

6. Describe how you might use a spinner divided into 6 equal slices (see here) to simulate the null model where the subjects are selecting at random. Include enough detail so that someone could read your description and know all of the steps to carry out to make the analysis. (*Hint: What would the numbers on the spinner represent? What does each spin represent? How many spins? Etc.*) (I am asking you to explain this procedure to help you think through what is going on in the One-Proportion applet.)

Create a spinner with 6 unlabeled blended meat products, spin the wheel a total of 18 times, one for each contestant choosing their least favorite dish. Write down the outcome of each spin to determine the totals for each product.

7. We will use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the dog food study according to the **null model** proposed by the skeptic, where the subjects are selecting at random.

7a. What will you use for the probability of success in this situation?

.1666

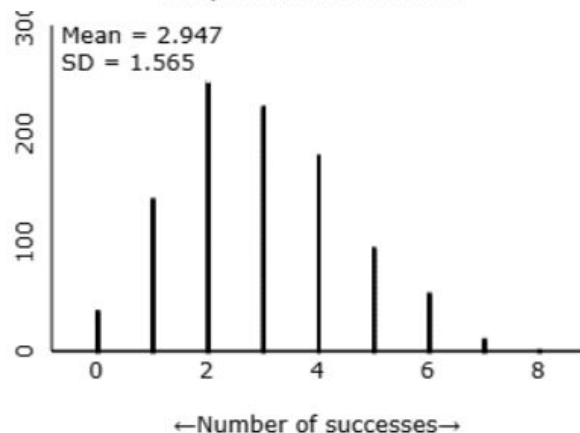
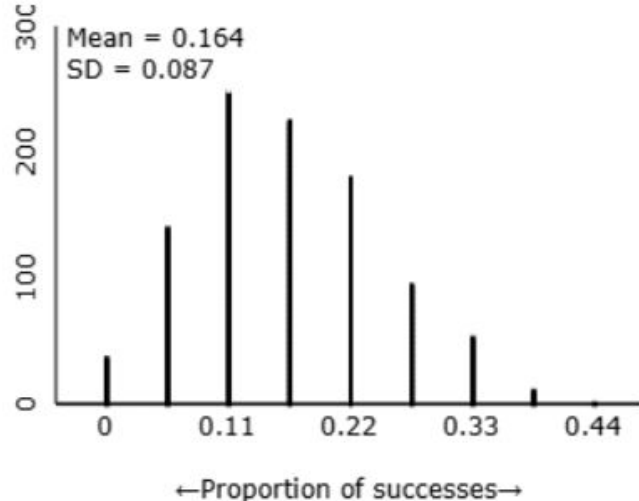
7b. What will you put in for the number of tosses?

18

7c. How many repetitions?

1000

8. Perform the simulation. Copy and paste the histogram you get here.



9. Describe what is going on in the graph in the usual way.

10. What conclusions can you draw from your simulation? As part of your answer, make sure to calculate an appropriate p-value and use it to support your conclusions.

P value = 0, so we reject the null hypothesis. Strong evidence that the dog food tasted the worst.

Group 4

Dog Food. An article in the February 20, 2009 issue of *Science* described a (lighthearted and informal) study in which 18 subjects (the author's New Year's Eve guests) were presented with six unlabeled blended meat products. One of these products was *Newman's Own* dog food, prepared with a food processor to have the texture and appearance of a liver mousse, and the other five were various gourmet mousse products. After tasting all six products, subjects were asked to identify which was their least favorite on the basis of taste. The research question is whether subjects are more likely to identify dog food as their least favorite than if they were making selections at random.

1. Who or what are the cases?

The 18 subjects.

2. What variable was recorded in this "study"? Is it categorical or numerical?

Least favorite product - categorical

3. Will your test involve a mean or a proportion? What notation will you use in the hypothesis test?

Proportion, $p_{\text{dog food}}$

It turns out that 13 out of 18 of the study subjects selected the dog food as their least favorite.

We are interested in testing the skeptic's theory that the subjects were in fact selecting their "least favorite" at random.

4. What proportion of the time would you expect the dog food to be selected if indeed the subjects were selecting at random? Calculate the proportion as a decimal.

$$1/6 = 0.167$$

5. Write out your null and alternate hypotheses, both in words and using correct notation.

H_0 : $P_{\text{dog food}} = .167$ - The dog food had no effect on which was their least favorite

H_a : $P_{\text{dog food}} > .167$ - The dog food affected which product was their least favorite

6. Describe how you might use a spinner divided into 6 equal slices (see here) to simulate the null model where the subjects are selecting at random. Include enough detail so that someone could read your description and know all of the steps to carry out to make the analysis. (*Hint: What would the numbers on the spinner represent? What does each spin represent? How many spins? Etc.*) (I am asking you to explain this procedure to help you think through what is going on in the One-Proportion applet.)

Each number on the spinner is one of the products. After selecting one of the numbers to be the dog food, we spin the spinner 18 times and mark how many times the dog food was selected.

7. We will use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the dog food study according to the **null model** proposed by the skeptic, where the subjects are selecting at random.

7a. What will you use for the probability of success in this situation?

.167

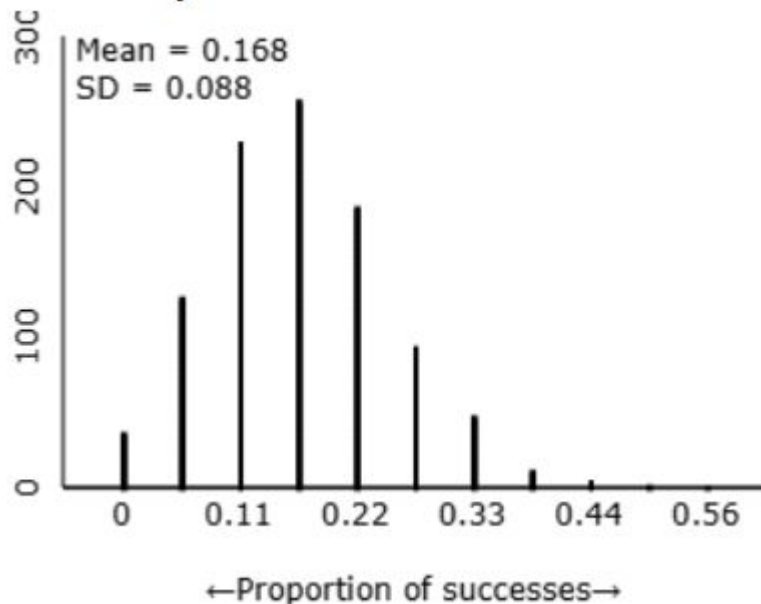
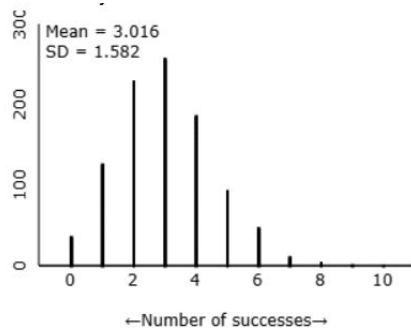
7b. What will you put in for the number of tosses?

18

7c. How many repetitions?

1001

8. Perform the simulation. Copy and paste the histogram you get here.



9. Describe what is going on in the graph in the usual way.

This is a graph of the proportions from 1001 simulated experiments. From each experiment, the proportion of guests who chose dog food as the worst product out of 18 guests. The proportions follow a normal distribution. The typical proportion was .168, give or take .088.

10. What conclusions can you draw from your simulation? As part of your answer, make sure to calculate an appropriate p-value and use it to support your conclusions.

p-value= 0

In 1001 simulated experiments, 0 had results as extreme as this one, resulting in a p-value of 0. Therefore, the probability of this result occurring due to chance is very small, meaning we have evidence to reject the null hypothesis.

Group 5

Dog Food. An article in the February 20, 2009 issue of *Science* described a (lighthearted and informal) study in which 18 subjects (the author's New Year's Eve guests) were presented with six unlabeled blended meat products. One of these products was *Newman's Own* dog food, prepared with a food processor to have the texture and appearance of a liver mousse, and the other five were various gourmet mousse products. After tasting all six products, subjects were asked to identify which was their least favorite on the basis of taste. The research question is whether subjects are more likely to identify dog food as their least favorite than if they were making selections at random.

1. Who or what are the cases?

The 18 subjects, the author's New Year's Eve guests

2. What variable was recorded in this "study"? Is it categorical or numerical?

Which product was their least favorite tastewise, categorical

3. Will your test involve a mean or a proportion? What notation will you use in the hypothesis test?

This will involve a proportion, and will be notated with P

It turns out that 13 out of 18 of the study subjects selected the dog food as their least favorite.

We are interested in testing the skeptic's theory that the subjects were in fact selecting their "least favorite" at random.

4. What proportion of the time would you expect the dog food to be selected if indeed the subjects were selecting at random? Calculate the proportion as a decimal.

$$\frac{1}{6} = 0.167$$

5. Write out your null and alternate hypotheses, both in words and using correct notation.

$$H_0: P_{\text{dogfood}} = 0.167$$

The subjects are no more likely to pick dog food compared to the other mousses as their least favorite

$$H_a: P_{\text{dogfood}} > 0.167$$

The subjects are likely to pick dog food as their least favorite mousse

6. Describe how you might use a spinner divided into 6 equal slices (see here) to simulate the null model where the subjects are selecting at random. Include enough detail so that someone could read your description and know all of the steps to carry out to make the analysis. (*Hint: What would the numbers on the spinner represent? What does each spin represent? How many spins? Etc.*) (I am asking you to explain this procedure to help you think through what is going on in the One-Proportion applet.)

The numbers on the spinner correspond with the six different blended meat products. We spin the spinner eighteen times to simulate the eighteen participants, and each time it lands signifies the least favorite product.

7. We will use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the dog food study according to the **null model** proposed by the skeptic, where the subjects are selecting at random.

7a. What will you use for the probability of success in this situation?

0.167

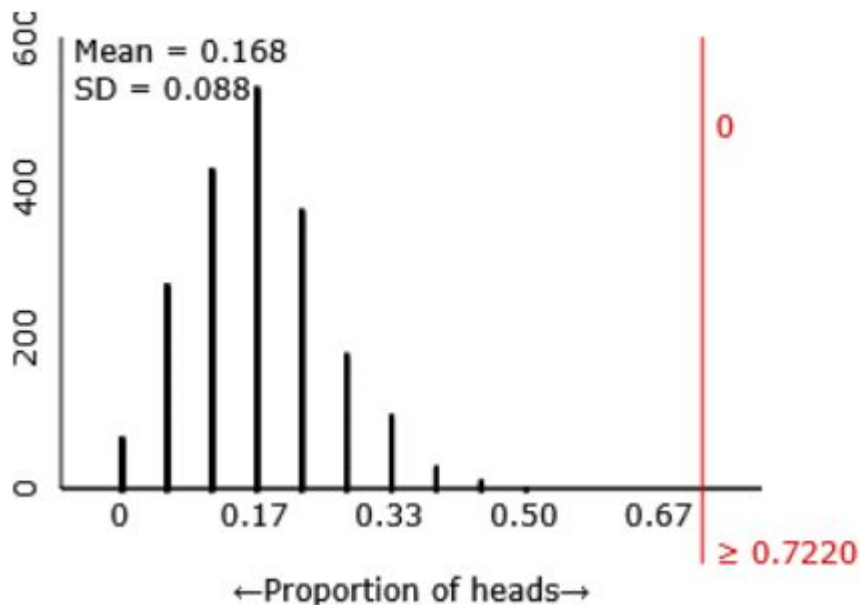
7b. What will you put in for the number of tosses?

18

7c. How many repetitions?

2000

8. Perform the simulation. Copy and paste the histogram you get here.



9. Describe what is going on in the graph in the usual way.

The graph represents the results of 2000 simulations of our experiment under the null hypothesis, showing the proportion of participants that selected dog food as their least favorite.

The data is normal, with a mean proportion of 0.168 and a standard deviation of 0.088, with most of the data falling roughly between 0.08 and 0.26

10. What conclusions can you draw from your simulation? As part of your answer, make sure to calculate an appropriate p-value and use it to support your conclusions.

Since 13/18 (0.72) participants selected dog food as their least favorite, and under the null hypothesis we observe any values equal to or greater than. The simulation has 0 instances of this happening, making its P-Value 0. Because of this, the result we got is extremely unlikely to be random chance, so we reject the null hypothesis

Group 6

Dog Food. An article in the February 20, 2009 issue of *Science* described a (lighthearted and informal) study in which 18 subjects (the author's New Year's Eve guests) were presented with six unlabeled blended meat products. One of these products was *Newman's Own* dog food, prepared with a food processor to have the texture and appearance of a liver mousse, and the other five were various gourmet mousse products. After tasting all six products, subjects were asked to identify which was their least favorite on the basis of taste. The research question is whether subjects are more likely to identify dog food as their least favorite than if they were making selections at random.

1. Who or what are the cases?
 - The 18 subjects invited to the meal
2. What variable was recorded in this “study”? Is it categorical or numerical?
 - Least Favorite food item on basis of taste - Categorical
3. Will your test involve a mean or a proportion? What notation will you use in the hypothesis test?
 - Proportion - number of times a food was picked as a least favorite food
 - p

Malec: $\bar{x}$ / μ are for averages. You (correctly!) stated this is about a proportion.

It turns out that 13 out of 18 of the study subjects selected the dog food as their least favorite.

We are interested in testing the skeptic's theory that the subjects were in fact selecting their "least favorite" at random.

4. What proportion of the time would you expect the dog food to be selected if indeed the subjects were selecting at random? Calculate the proportion as a decimal.

If the subjects least favorite was selected at random I would expect the dog food to have been picked roughly 1/6th of the time. With 18 subjects and being picked randomly 1/6th of the time the dog food should have been picked roughly 3 times by the 18 subjects.

5. Write out your null and alternative hypotheses, both in words and using correct notation.

H₀: The dog food should have the same random chance of being picked as any of the other foods.

$$H_0: p = 1/6$$

H_a: The dog food was not picked by chance amongst the subjects and so should have a p value greater than 1/6th.

$$H_a: p > 1/6$$

P = the proportion of guests who think the dog food is the worst.

6. Describe how you might use a spinner divided into 6 equal slices (see here) to simulate the null model where the subjects are selecting at random. Include enough detail so that someone could read your description and know all of the steps to carry out to make the analysis. (*Hint: What would the numbers on the spinner represent? What does each spin represent? How many spins? Etc.*) (I am asking you to explain this procedure to help you think through what is going on in the One-Proportion applet.)

The spinner will have six equal wedges marked out with a number corresponding to the food variables. Each spin will be a subject picking that particular number to be their least favorite food for the test. For each simulated experiment the spinner should be spun 18 times, once for each of the 18 subjects.

7. We will use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the dog food study according to the **null model** proposed by the skeptic, where the subjects are selecting at random.

7a. What will you use for the probability of success in this situation?

16.666667%

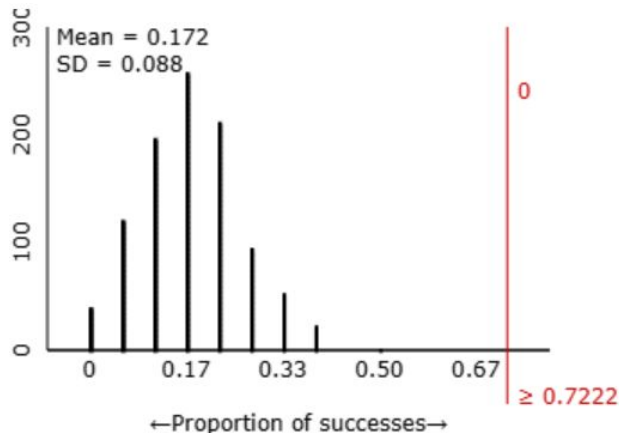
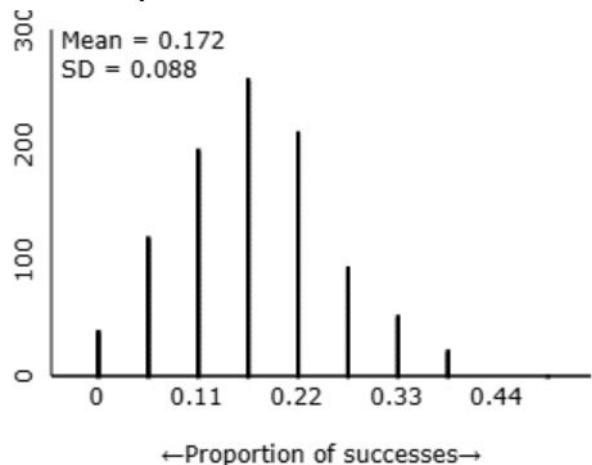
7b. What will you put in for the number of tosses?

18

7c. How many repetitions?

1000

8. Perform the simulation. Copy and paste the histogram you get here.



9. Describe what is going on in the graph in the usual way.

The graph shows a normal bell curve distribution with a center of 0.172. This means that the average study assuming the null hypothesis only chose the dog food as their least favorite 17.2% of the time.

10. What conclusions can you draw from your simulation? As part of your answer, make sure to calculate an appropriate p-value and use it to support your conclusions.

There was zero simulations that had achieved similar results to the original study's results. This would make our p-value equal 0 meaning that we can reject the null hypothesis.

Group 7

Dog Food. An article in the February 20, 2009 issue of *Science* described a (lighthearted and informal) study in which 18 subjects (the author's New Year's Eve guests) were presented with six unlabeled blended meat products. One of these products was *Newman's Own* dog food, prepared with a food processor to have the texture and appearance of a liver mousse, and the other five were various gourmet mousse products. After tasting all six products, subjects were asked to identify which was their least favorite on the basis of taste. The research question is whether subjects are more likely to identify dog food as their least favorite than if they were making selections at random.

1. Who or what are the cases?

The 18 New Year's Eve guests

2. What variable was recorded in this "study"? Is it categorical or numerical?

Which food will identify their least favorite meat blend.

Categorical

3. Will your test involve a mean or a proportion? What notation will you use in the hypothesis test?

Population Proportion. P

It turns out that 13 out of 18 of the study subjects selected the dog food as their least favorite.

We are interested in testing the skeptic's theory that the subjects were in fact selecting their "least favorite" at random.

4. What proportion of the time would you expect the dog food to be selected if indeed the subjects were selecting at random? Calculate the proportion as a decimal.

0.16666

5. Write out your null and alternative hypotheses, both in words and using correct notation.

H_0 : The subjects (P) will select each type of meat blend for their least liked at an even proportion $p = 1/6$

H_a : The subjects (P) will select dog food for their least liked at an uneven proportion. $p \neq 1/6$

Malec: Be sure to write this in notation too

6. Describe how you might use a spinner divided into 6 equal slices (see here) to simulate the null model where the subjects are selecting at random. Include enough detail so that someone could read your description and know all of the steps to carry out to make the analysis. (*Hint: What would the numbers on the spinner represent? What does each spin represent? How many spins? Etc.*) (I am asking you to explain this procedure to help you think through what is going on in the One-Proportion applet.)

1. Have a spinner in the center of a circle with 6 equal possibilities
2. Spin the spinner 18 times
3. If the spinner lands at:
 - a. 1: meat 1
 - b. 2: meat 2
 - c. 3: meat 3
 - d. 4: meat 4
 - e. 5: meat 5
 - f. 6: dog food
4. Each spin represents a person and their choice of food
7. We will use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the dog food study according to the **null model** proposed by the skeptic, where the subjects are selecting at random.

7a. What will you use for the probability of success in this situation?

0.166667

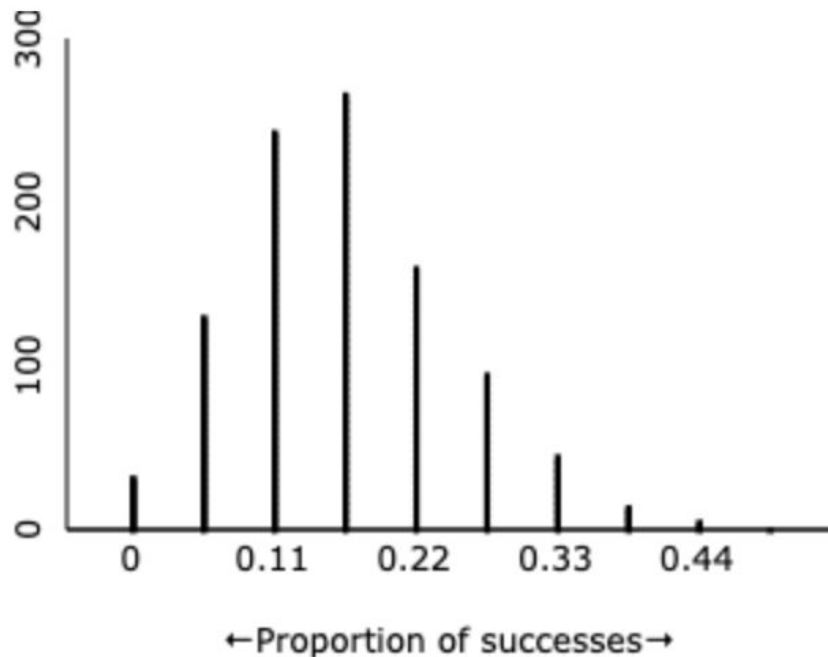
7b. What will you put in for the number of tosses?

18

7c. How many repetitions?

1000

8. Perform the simulation. Copy and paste the histogram you get here.



9. Describe what is going on in the graph in the usual way.

The graph is a description of how many people would pick dog food as their least favorite food under the assumption that people would randomly pick their least favorite. The graph has a mean of 0.1666, indicating that 16% of the time, people would randomly pick dog food as their least favorite. The graph has a range of 0 - 0.50.

10. What conclusions can you draw from your simulation? As part of your answer, make sure to calculate an appropriate p-value and use it to support your conclusions.

From our simulation, we can determine that the people selected dog food at an ____, higher than the p value of 0.02.

Group 8

Dog Food. An article in the February 20, 2009 issue of *Science* described a (lighthearted and informal) study in which 18 subjects (the author's New Year's Eve guests) were presented with six unlabeled blended meat products. One of these products was *Newman's Own* dog food, prepared with a food processor to have the texture and appearance of a liver mousse, and the other five were various gourmet mousse products. After tasting all six products, subjects were asked to identify which was their least favorite on the basis of taste. The research question is whether subjects are more likely to identify dog food as their least favorite than if they were making selections at random.

1. Who or what are the cases?

The cases are the guests that were served the food

2. What variable was recorded in this “study”? Is it categorical or numerical?

The variable that was recorded in the study was a categorical variable about which product the subject of the experiment liked the least

3. Will your test involve a mean or a proportion? What notation will you use in the hypothesis test?

The test will involve a proportion as opposed to a mean because the variables are categorical and not numerical

It turns out that 13 out of 18 of the study subjects selected the dog food as their least favorite.

We are interested in testing the skeptic's theory that the subjects were in fact selecting their "least favorite" at random.

4. What proportion of the time would you expect the dog food to be selected if indeed the subjects were selecting at random? Calculate the proportion as a decimal.

$$\frac{1}{6} \times \frac{3}{3} = \frac{3}{18} = 0.16667$$

5. Write out your null and alternate hypotheses, both in words and using correct notation.

H_0 : the proportion of 13/18 or 0.72222 was randomly selected and occurred due to random chance

H_a : the proportion of 0.72222 that was found in the study was not randomly selected and there was a relevant variable which the study is not directly accounting for.

6. Describe how you might use a spinner divided into 6 equal slices (see here) to simulate the null model where the subjects are selecting at random. Include enough detail so that someone could read your description and know all of the steps to carry out to make the analysis. (*Hint: What would the numbers on the spinner represent? What does each spin represent? How many spins? Etc.*) (I am asking you to explain this procedure to help you think through what is going on in the One-Proportion applet.)

To simulate the null model I would create a spinner with 6 equal parts and label one of the parts as the dog food. I would then spin it 18 times and record how many times it landed on the dog food. I would then find the proportion of that and record it. I would do this as many times as needed to achieve precise results. In this situation the numbers on the spinner would represent the different types of food served to the guests and each spin represents a case in the study.

7. We will use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the dog food study according to the **null model** proposed by the skeptic, where the subjects are selecting at random.

7a. What will you use for the probability of success in this situation?

0.166667

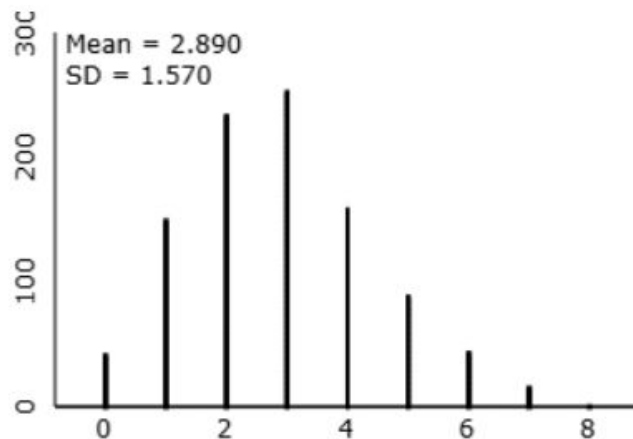
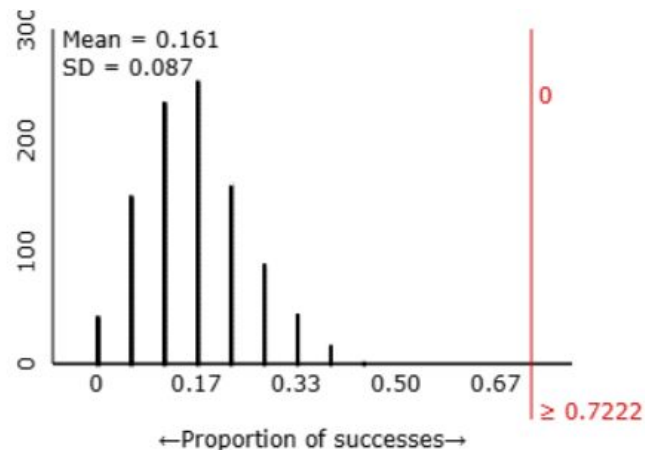
7b. What will you put in for the number of tosses?

18

7c. How many repetitions?

1000

8. Perform the simulation. Copy and paste the histogram you get here.



9. Describe what is going on in the graph in the usual way.

The graph has a slightly left leaning bell curve. It has a mean of 2.89 and a standard deviation of 1.57.

10. What conclusions can you draw from your simulation? As part of your answer, make sure to calculate an appropriate p-value and use it to support your conclusions.

From my simulation I can conclude that the null hypothesis is likely incorrect because the chance of randomly getting a proportion of 0.7222 is immensely low

Group 9

Dog Food. An article in the February 20, 2009 issue of *Science* described a (lighthearted and informal) study in which 18 subjects (the author's New Year's Eve guests) were presented with six unlabeled blended meat products. One of these products was *Newman's Own* dog food, prepared with a food processor to have the texture and appearance of a liver mousse, and the other five were various gourmet mousse products. After tasting all six products, subjects were asked to identify which was their least favorite on the basis of taste. The research question is whether subjects are more likely to identify dog food as their least favorite than if they were making selections at random.

1. Who or what are the cases?

The authors 18 new years eves guests

2. What variable was recorded in this "study"? Is it categorical or numerical?

Identifying the dog food, categorical

3. Will your test involve a mean or a proportion? What notation will you use in the hypothesis test?

Proportion, P

It turns out that 13 out of 18 of the study subjects selected the dog food as their least favorite.

We are interested in testing the skeptic's theory that the subjects were in fact selecting their "least favorite" at random.

4. What proportion of the time would you expect the dog food to be selected if indeed the subjects were selecting at random? Calculate the proportion as a decimal.

$\frac{1}{6}$ times = .166

5. Write out your null and alternate hypotheses, both in words and using correct notation.

H_0 : $p = .167$ the subjects will choose the dog food one out of 6 people every time

H_a : $p > .167$ the subjects will choose the dog food more than 1 out of 6 people every time

6. Describe how you might use a spinner divided into 6 equal slices (see here) to simulate the null model where the subjects are selecting at random. Include enough detail so that someone could read your description and know all of the steps to carry out to make the analysis. (*Hint: What would the numbers on the spinner represent? What does each spin represent? How many spins? Etc.*) (I am asking you to explain this procedure to help you think through what is going on in the One-Proportion applet.)

Put numbers on the spinner, each number represents 1 of the 6 food items. Each spin represents a choice, and there can only be 18 spins. To make it even more randomized the places of the food can be switched around after each spin.

7. We will use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the dog food study according to the **null model** proposed by the skeptic, where the subjects are selecting at random.

7a. What will you use for the probability of success in this situation?

.167 because its 1 out of every 6 times

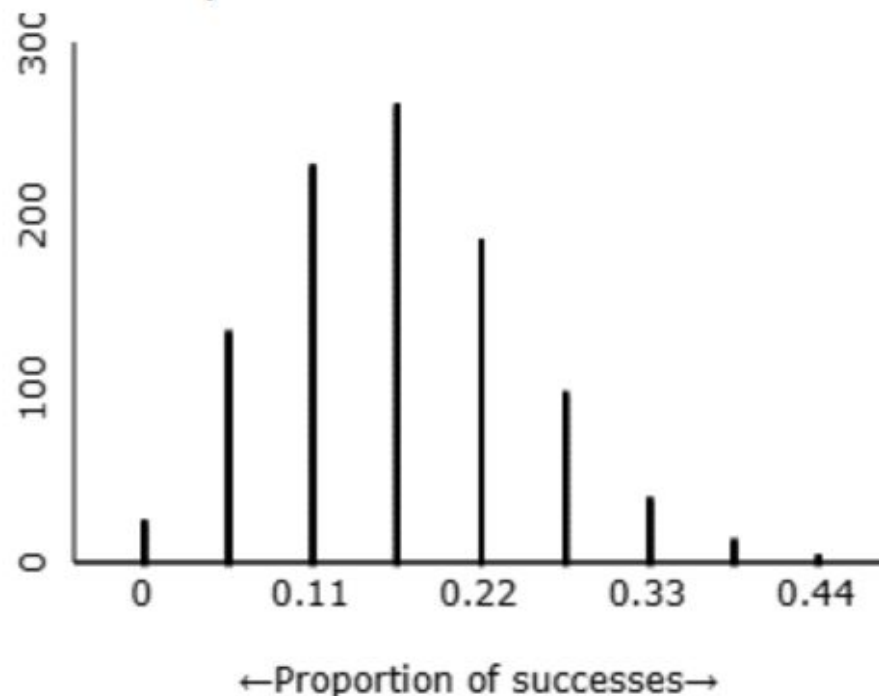
7b. What will you put in for the number of tosses?

18

7c. How many repetitions?

1000

8. Perform the simulation. Copy and paste the histogram you get here.



9. Describe what is going on in the graph in the usual way.

The graph shows the data after 1000 samples were ran and also proportion of successes. The graph has a bell shaped curve and the mean is around .167 which could also be considered the middle.

10. What conclusions can you draw from your simulation? As part of your answer, make sure to calculate an appropriate p-value and use it to support your conclusions.

We can see that in the simulation there is 0 simulations that come up with the number of .7222, so we can reject the null hypothesis

Group 10

Dog Food. An article in the February 20, 2009 issue of *Science* described a (lighthearted and informal) study in which 18 subjects (the author's New Year's Eve guests) were presented with six unlabeled blended meat products. One of these products was *Newman's Own* dog food, prepared with a food processor to have the texture and appearance of a liver mousse, and the other five were various gourmet mousse products. After tasting all six products, subjects were asked to identify which was their least favorite on the basis of taste. The research question is whether subjects are more likely to identify dog food as their least favorite than if they were making selections at random.

1. Who or what are the cases?

The people who were fed the food

2. What variable was recorded in this "study"? Is it categorical or numerical?

What meat they hated the most, categorical

3. Will your test involve a mean or a proportion? What notation will you use in the hypothesis test?

I think a proportion will be best here, because it's the proportion of the ones who choose dog food.

It turns out that 13 out of 18 of the study subjects selected the dog food as their least favorite.

We are interested in testing the skeptic's theory that the subjects were in fact selecting their "least favorite" at random.

4. What proportion of the time would you expect the dog food to be selected if indeed the subjects were selecting at random? Calculate the proportion as a decimal.

I'd say that out of every six people, at least 1 should be picking dog food, so the proportion would be .222

Malec: check your arithmetic here - $\frac{1}{6}$ isn't .1875

5. Write out your null and alternate hypotheses, both in words and using correct notation.

H_0 : $m = .222$ the subjects will choose the dog food as their least favorite .1875 percent of the time

H_a : $m > .222$ the subjects are likely to choose the dog food as their least favorite more than .1875 percent of the time.

Malec: you (correctly) said this is a proportion on the last slide, so you should be using p . Where'd that .222 come from?

6. Describe how you might use a spinner divided into 6 equal slices (see here) to simulate the null model where the subjects are selecting at random. Include enough detail so that someone could read your description and know all of the steps to carry out to make the analysis. (*Hint: What would the numbers on the spinner represent? What does each spin represent? How many spins? Etc.*) (I am asking you to explain this procedure to help you think through what is going on in the One-Proportion applet.)

The spinner will be spun a total of 16 times, each spin representing a person in each of the trials, each slice having an equal chance of being picked as the others, with each number representing a different meat type, and the results of each spin are recorded. The proportion of the spins that didn't get the dog food and then the dog food is then calculated to find the proportion. This is then repeated 1000 time to repeat the sample process.

7. We will use the One-Proportion applet (at the link <http://www.rossmanchance.com/applets/OneProp/OneProp.htm>) to simulate at least 1000 repetitions of the dog food study according to the **null model** proposed by the skeptic, where the subjects are selecting at random.

7a. What will you use for the probability of success in this situation?

.16666667

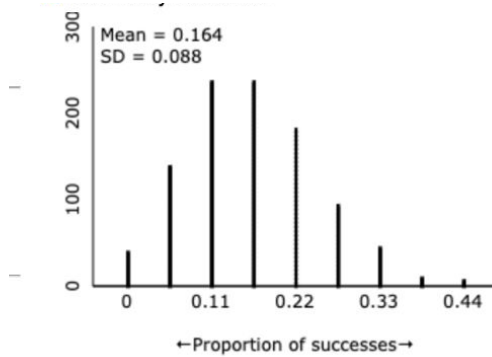
7b. What will you put in for the number of tosses?

18

7c. How many repetitions?

1000

8. Perform the simulation. Copy and paste the histogram you get here.



9. Describe what is going on in the graph in the usual way. The graph is a depiction of the counts of proportions of successes assuming the null hypothesis is true, per 1000 samples. The graph is a normal bell curve graph, with an average datapoint being .164, with a standard distance of .088 between points.

10. What conclusions can you draw from your simulation? As part of your answer, make sure to calculate an appropriate p-value and use it to support your conclusions.

Based on the graph, it seems as though the possibility of getting the alternate hypothesis of .7222 is practically impossible when assuming the null hypothesis is true, as there is not a single point even approaching that proportion on the graph. Thus, this makes it very likely that this study rejects the null hypothesis